KAIZEN MATHS

Lesson starter

HCF, LCM and Prime Factorisation

Mixed Problems

Name _____

S1

LCM(14, 21)

S2

HCF(45, 35)

S3

LCM(24, 18)

S4

LCM(12, 9)

Show your working

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NUM-A60B-L3 · S

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KAIZEN MATHS

Exit ticket

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Name _____

E1

HCF(12, 30)

E2

HCF(33, 22)

E3

HCF(28, 12)

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Exit ticket

HCF, LCM and Prime Factorisation

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E2

HCF(33, 22)

E3

HCF(28, 12)

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NUM-A60B-L3 · E

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NUM-A60B-L3 · E

Show your working

Teacher answers & working

HCF, LCM and Prime Factorisation · Mixed Problems

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S1

LCM(14, 21)

Answer:

42

Working:

Find the LCM of 14 and 21

Prime factorisation of 14: 2×7

Prime factorisation of 21: 3×7

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2 \times 3 \times 7$

Multiply these factors: $2 \times 3 \times 7 = 42$

$\text{LCM} = 42$

S2

HCF(45, 35)

Answer:

5

Working:

Find the HCF of 45 and 35

Prime factorisation of 45: $3^2 \times 5$

Prime factorisation of 35: 5×7

Common prime factors: 5

Multiply the common prime factors: $5 = 5$

$\text{HCF} = 5$

Teacher answers & working

HCF, LCM and Prime Factorisation · Mixed Problems

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S3

LCM(24, 18)

Answer:

72

Working:

Find the LCM of 24 and 18

Prime factorisation of 24: $2^3 \times 3$

Prime factorisation of 18: 2×3^2

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2^3 \times 3^2$

Multiply these factors: $2^3 \times 3^2 = 72$

LCM = 72

S4

LCM(12, 9)

Answer:

36

Working:

Find the LCM of 12 and 9

Prime factorisation of 12: $2^2 \times 3$

Prime factorisation of 9: 3^2

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2^2 \times 3^2$

Multiply these factors: $2^2 \times 3^2 = 36$

LCM = 36

Teacher answers & working

HCF, LCM and Prime Factorisation · Mixed Problems

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E1

HCF(12, 30)

Answer:

6

Working:

Find the HCF of 12 and 30

Prime factorisation of 12: $2^2 \times 3$

Prime factorisation of 30: $2 \times 3 \times 5$

Common prime factors: 2×3

Multiply the common prime factors: $2 \times 3 = 6$

HCF = 6

E2

HCF(33, 22)

Answer:

11

Working:

Find the HCF of 33 and 22

Prime factorisation of 33: 3×11

Prime factorisation of 22: 2×11

Common prime factors: 11

Multiply the common prime factors: $11 = 11$

HCF = 11

E3

HCF(28, 12)

Answer:

4

Working:

Find the HCF of 28 and 12

Prime factorisation of 28: $2^2 \times 7$

Prime factorisation of 12: $2^2 \times 3$

Common prime factors: 2^2

Multiply the common prime factors: $2^2 = 4$

HCF = 4